

MINI PROJECT ON DEVELOPMENT FOR STUDENT RESULT ANALYZER

**Submitted in the partial fulfilment of the requirements for the
award of degree of**

BACHELORS IN COMPUTER APPLICATIONS

BCA

TO

UTTARANCHAL SCHOOL OF COMPUTING SCIENCES

UTTARANCHAL UNIVERSITY, DEHRADUN



**UTTARANCHAL
UNIVERSITY**

Under the supervision of:

Mr. Aagman Kaparwan

(Assistant Professor)

Submitted By:

Aditya Khatri

UU2509000023

UTTARANCHAL UNIVERSITY, (DEHRADUN) 2026

ACKNOWLEDGEMENT

First and foremost, I will be eternally grateful to God, to whom I owe me life. I'd also like to thank my parents for allowing me to attend Uttaranchal University in Dehradun. I'd like to convey my heartfelt appreciation to our Dean, Prof. (Dr.) Sonal Sharma, and HOD Sir, Dr. Sameer Dev Sharma, for their valuable support and guidance. I'd like to convey my heartfelt appreciation to us Project Mentors, Assistant Prof. Aagman Kaparwan for their invaluable assistance in preparing the project and compiling the project materials. I am grateful for their consistent blessings and for providing the necessary and related facilities for our computer project file. Most importantly. I'd also like to thank everyone who helped me finish my project, whether directly or indirectly.

Aditya Khatri
UU2509000023
BCA 1'E'

DECLARATION

I hereby declare that this project work entitled “**STUDENT RESULT ANALYZER**” submitted by **Aditya Khatri to Uttarakhand Institute of Computing Sciences**. It is an authentic record of my work carried out at “**UTTARANCHAL UNIVERSITY**” under the guidance of Prof. Aagman Kaparwan.

Aditya Khatri
UU2509000023
BCA 1'E'

CERTIFICATE OF ORIGINALITY



**UTTARANCHAL
UNIVERSITY**

This is to certify that the project entitled “**STUDENT RESULT ANALZER**” by **ADITYA KHATRI** have been submitted in partial fulfilment of the requirements for the award of the degree of BCA-I from Uttaranchal University, Dehradun. The results embodied in this project have not been submitted to any other University or Institution for the record of any degree.

Under the guidance of:

Mr. Aagman Kaparwan

Assistant professor (USCS)

Uttaranchal University, Dehradun

TABLE OF CONTENTS

ACKNOWLEDGEMENT	I
DECLARATION	II
CERTIFICATE OF ORIGINALITY	III
1.INTRODUCTION.....	1
2. OBJECTIVE	2
1.HARDWARE AND SOFTWARE USED	3
3.1 Software requirement	3
3.2 Hardware Requirement	3
4. SYSTEM ANALYSIS	4
4.1 IDENTIFICATION OF NEED	4
4.1.2 PROPOSED SYSTEM.....	5
5. PRELIMINARY INVESTIGATION.....	6
5.1 FEASIBILITY STUDY.....	6
6. PROJECT PLANNING	7
7. PROJECT SCHEDULING	8
7.1 GANTT CHART	8
8. DATA MODELS.....	9
8.1 DFD (Data Flow Diagram).....	9
8.2 CLASS DIAGRAM	10
8.3 ACTIVITY DIAGRAM	11
9. METHODOLOGY	12
10. FUTURE SCOPE	13
11. IT COMPANIES.....	14
12. CODING	15
13. OUTPUT	20
14. BIBLIOGRAPHY	22

1. INTRODUCTION

A Student Result Analyzer is a useful system developed to make the process of managing and analysing student results easier and more organized. In many educational institutions, maintaining and calculating student results manually can take a lot of time and may also lead to mistakes. Teachers have to calculate total marks, percentages, and grades for many students, which can be difficult without a proper system. The Student Result Analyzer helps solve this problem by using a computer program to manage and process student result data efficiently.

This system allows the user to enter important details such as student name, roll number, and marks obtained in different subjects. After entering the data, the program can automatically calculate the total marks and percentage of each student. It can also determine whether a student has passed or failed based on the marks entered. By doing these calculations automatically, the system reduces human errors and saves time.

The Student Result Analyzer also helps in analysing student performance in a simple way. It can display the results of all students and sometimes identify the highest or lowest marks in the class. This makes it easier for teachers to understand the academic performance of students and helps in evaluating their progress.

Overall, the Student Result Analyzer improves the accuracy and speed of result management. It is a simple yet useful project for students who are learning programming with C++. By developing this project, BCA students can understand how programming concepts like classes, functions, loops, and conditional statements can be used to create a real-life application.

2. OBJECTIVE

The main objective of the Student Result Analyzer System is to efficiently manage and analyze the academic results of students using a computer-based program. In many educational institutions, calculating and maintaining student results manually can be time-consuming and may lead to errors. This system is designed to help teachers or users easily store student details and their marks in different subjects. By using C++ programming, the project aims to automate the process of calculating total marks, percentages, and grades, making result management faster, more accurate, and reliable.

The Student Result Analyzer helps in organizing student data in a structured way so that results can be viewed and analyzed easily. It reduces the need for manual calculations and paperwork, which saves time and improves efficiency. The system also helps in understanding the overall performance of students by displaying their results clearly. Through this project, students can learn how programming concepts can be applied to solve real-life academic data management problems.

The key objectives of this project are:

1. **To manage and analyze student results by automating** the calculation of total marks, percentage, and grades.
2. **To store and maintain student details** such as name, roll number, and marks obtained in different subjects.
3. **To reduce manual calculation errors and save time** by using a computerized result processing system.
4. **To help teachers or users easily view and analyze** the academic performance of students.
5. **To organize student result information** in a clear and structured format.
6. **To improve the efficiency and accuracy of result management** in educational institutions.
7. **To provide practical experience of using C++ programming concepts** like classes, functions, and loops in a real-life application.

1. HARDWARE AND SOFTWARE USED

3.1 Software requirement

- Programming Language: C++
- Compiler / IDE: Turbo C++ / Code::Blocks / Dev C++
- Operating System: Windows 7, Windows 8, Windows 10 or above

3.2 Hardware Requirement

Minimum Requirements

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB
- Hard Disk: Minimum 10 GB free disk space
- Keyboard and Mouse: For user input and navigation
- Monitor: Standard display monitor

4. SYSTEM ANALYSIS

System Analysis is the process of studying the existing system to understand its working and identify problems. It helps in designing a better system that improves efficiency, reduces errors, and manages student result data in a more organized and accurate way.

4.1 IDENTIFICATION OF NEED

In educational institutions, managing and analysing student results is an important task for teachers and academic staff. Student marks, grades, and performance records must be maintained properly to evaluate academic progress. However, when this process is done manually using registers or spreadsheets, it can become time-consuming and may lead to calculation errors. Mistakes in calculating total marks, percentages, or grades can affect the accuracy of student results.

Teachers often have to manage the results of many students at the same time. Calculating totals, percentages, and pass or fail status for each student manually requires a lot of effort and time. It also becomes difficult to quickly analyse overall class performance or identify students who performed very well or poorly. Such manual processes can reduce efficiency and make result management more complicated.

In today's digital era, there is a growing need for simple computer-based systems that can manage academic data more efficiently. A Student Result Analyzer helps in solving these problems by automating the process of storing student details and calculating their results. The system can quickly process marks, generate totals and percentages, and display the results in an organized format.

Therefore, the need for a Student Result Analyzer arises from the limitations of manual result management and the need for a faster, more accurate, and organized system. This project helps improve the efficiency of result processing and provides a better way to analyse student academic performance.

4.1.1 EXISTING SYSTEM

Since no one else site like that really appears to exist. As a result, no existing persists.

4.1.2 PROPOSED SYSTEM

The proposed Student Result Analyzer System is designed to make the process of managing and analysing student results easier, faster, and more accurate. It will replace the traditional manual method of calculating and recording student marks with a computerized system developed using C++. This system will help teachers or users quickly store student details, calculate results, and analyse academic performance.

- 1 **Student Record Management** – Store important student details such as name, roll number, and marks obtained in different subjects.
- 2 **Automatic Result Calculation** – Automatically calculate total marks and percentage of each student.
- 3 **Pass/Fail Identification** – Determine whether a student has passed or failed based on the marks entered.
- 4 **Result Display System** – Display the results of students in an organized and easy-to-read format.
- 5 **Performance Analysis** – Help in analysing student performance by comparing marks or identifying highest and lowest scores.
- 6 **Error Reduction** – Reduce manual calculation mistakes by performing automatic computations.
- 7 **Efficient Data Management** – Maintain student result data in a structured and organized way for quick access.

5. PRELIMINARY INVESTIGATION

The purpose of preliminary investigation is to study the existing system and identify the problems present in it. In the current system, student results are often calculated and recorded manually, which can take more time and may lead to errors. Therefore, this project investigates the need for a simple computerized system using C++ that can store student details and automatically calculate results more accurately and efficiently.

5.1 FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed project is practical and beneficial. It helps in analysing if the system can be developed with the available resources, technology, and time. For the Student Result Analyzer, the feasibility study ensures that the project can be implemented easily using basic programming concepts and tools.

Technically Feasibility:

The project is technically feasible because it uses the C++ programming language, which is commonly available and easy to use for developing small applications. The required tools such as compilers or IDEs like Code::Blocks or Dev C++ are freely available, and the technical skills required to develop this project are suitable for BCA students.

Operational Feasibility:

The proposed system is easy to operate and understand. It helps users store student details and automatically calculate results such as total marks and percentage. This reduces manual work and improves efficiency in managing student result data.

Economic Feasibility:

The project is economically feasible because it does not require expensive hardware or software. All the tools used for development are free and easily available, which makes the development and maintenance cost very low.

6. PROJECT PLANNING

Project planning is an important step that defines the overall vision and direction of the project. It explains how the project will be developed, implemented, and evaluated during the development process.

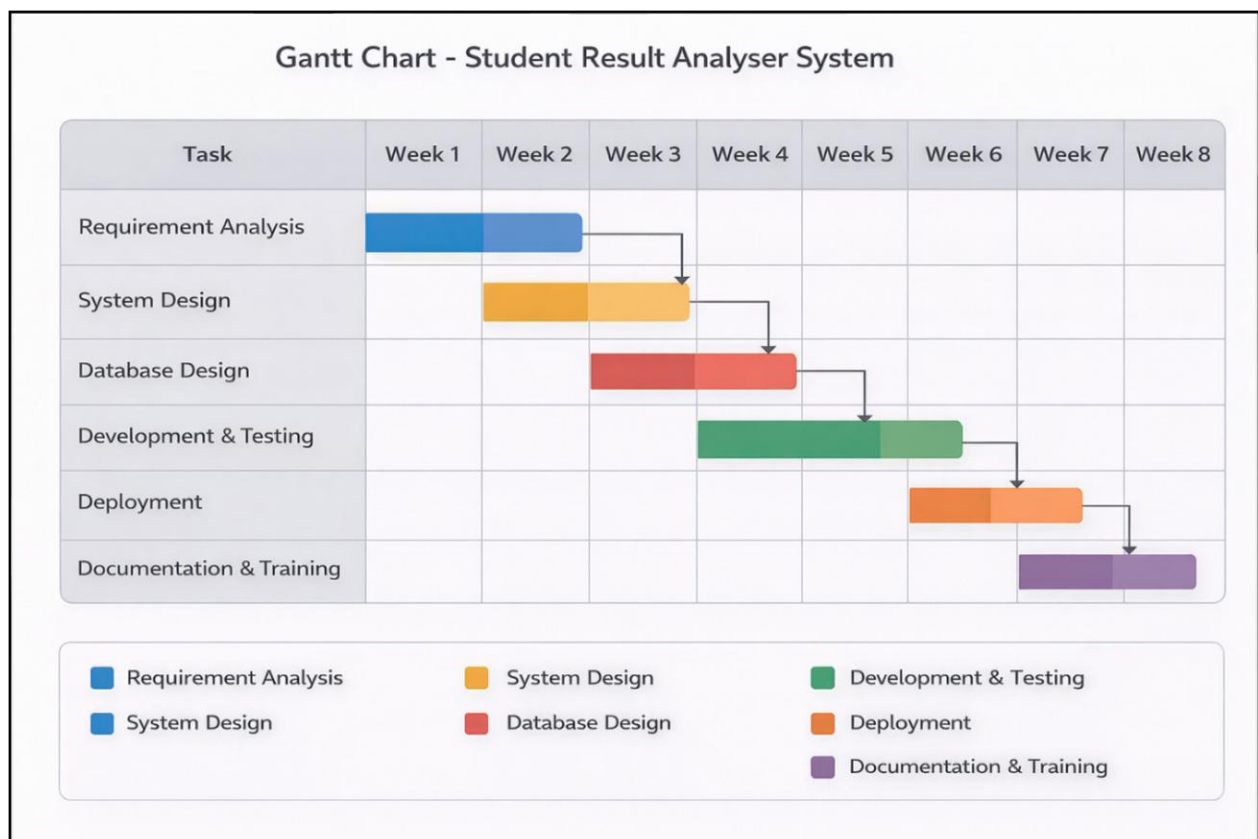
It includes the details of different steps required to complete the project, such as designing the program, writing the code, testing the system, and final implementation. It also helps in managing the time and resources required to complete the Student Result Analyzer project successfully.

7. PROJECT SCHEDULING

A project schedule is simply for conveying and understanding what activities are required & must be accomplished, what organizational resources will be provided or allocated to these tasks, and when the work must be completed. Cost savings, Successful project, & customer pleasure all benefits from effective project scheduling.

7.1 GANTT CHART

A Gantt chart is a graphical depiction of activities as they relate to time. It is used in project management to assist with project scheduling of all complex programming.

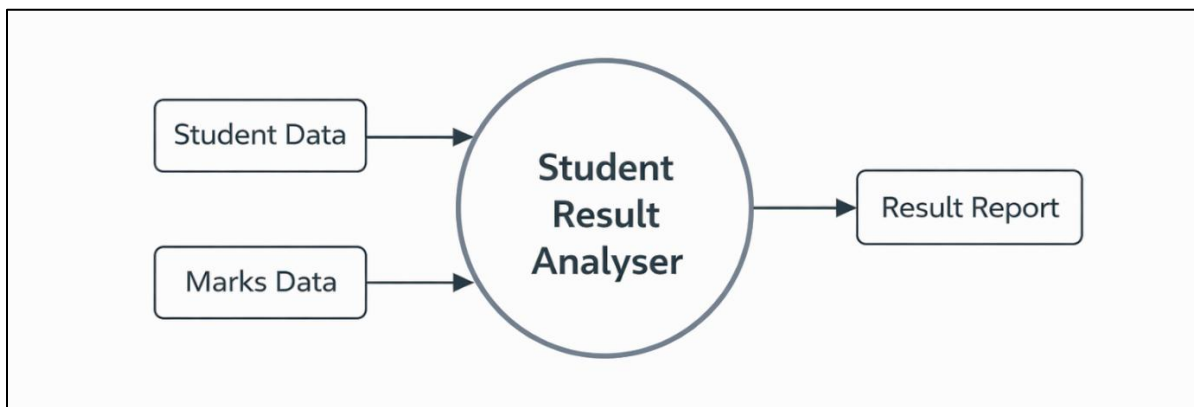


8. DATA MODELS

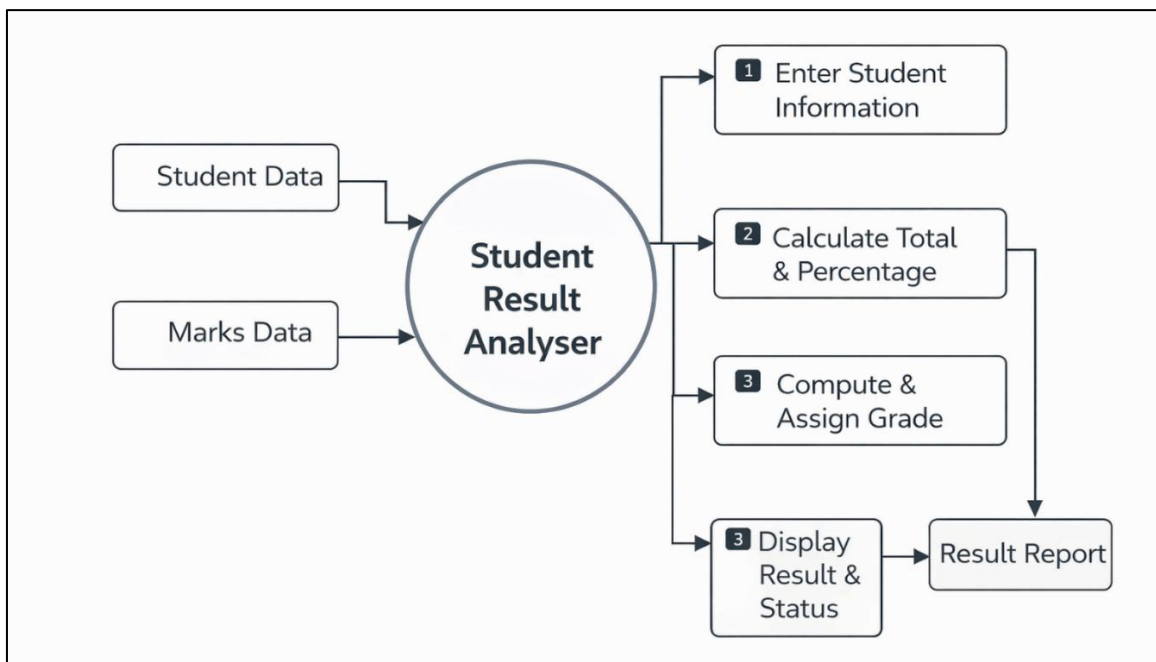
8.1 DFD (Data Flow Diagram)

A data flow diagram (DFD) is a graphical or visual representation of a business's operations through data movement that utilizes a recognized set of symbols and notations.

DFD 0 LEVEL



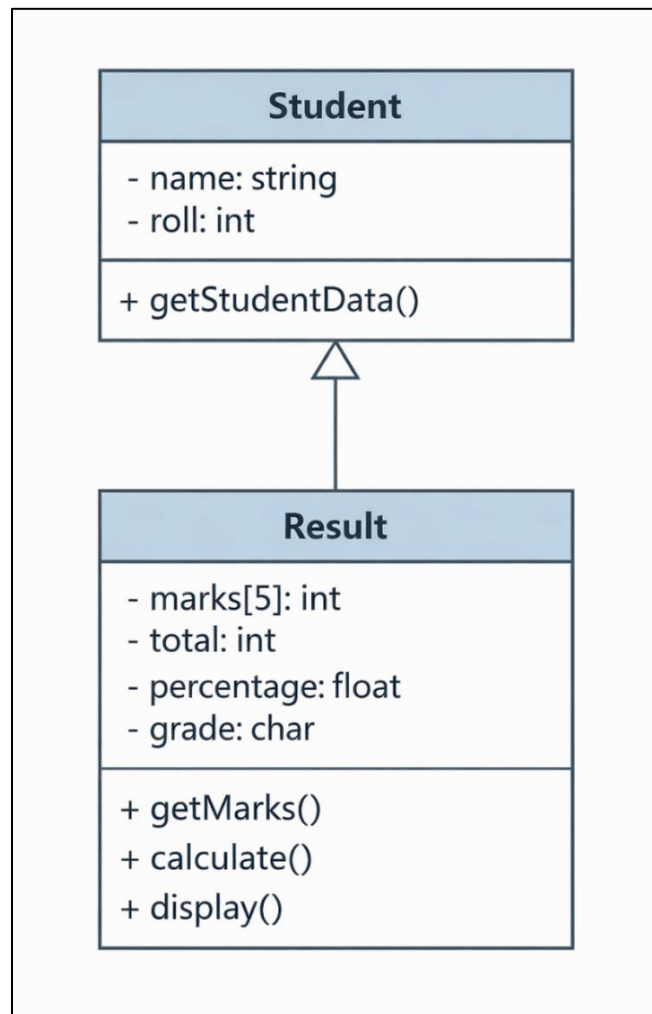
DFD 1 LEVEL



8.2 CLASS DIAGRAM

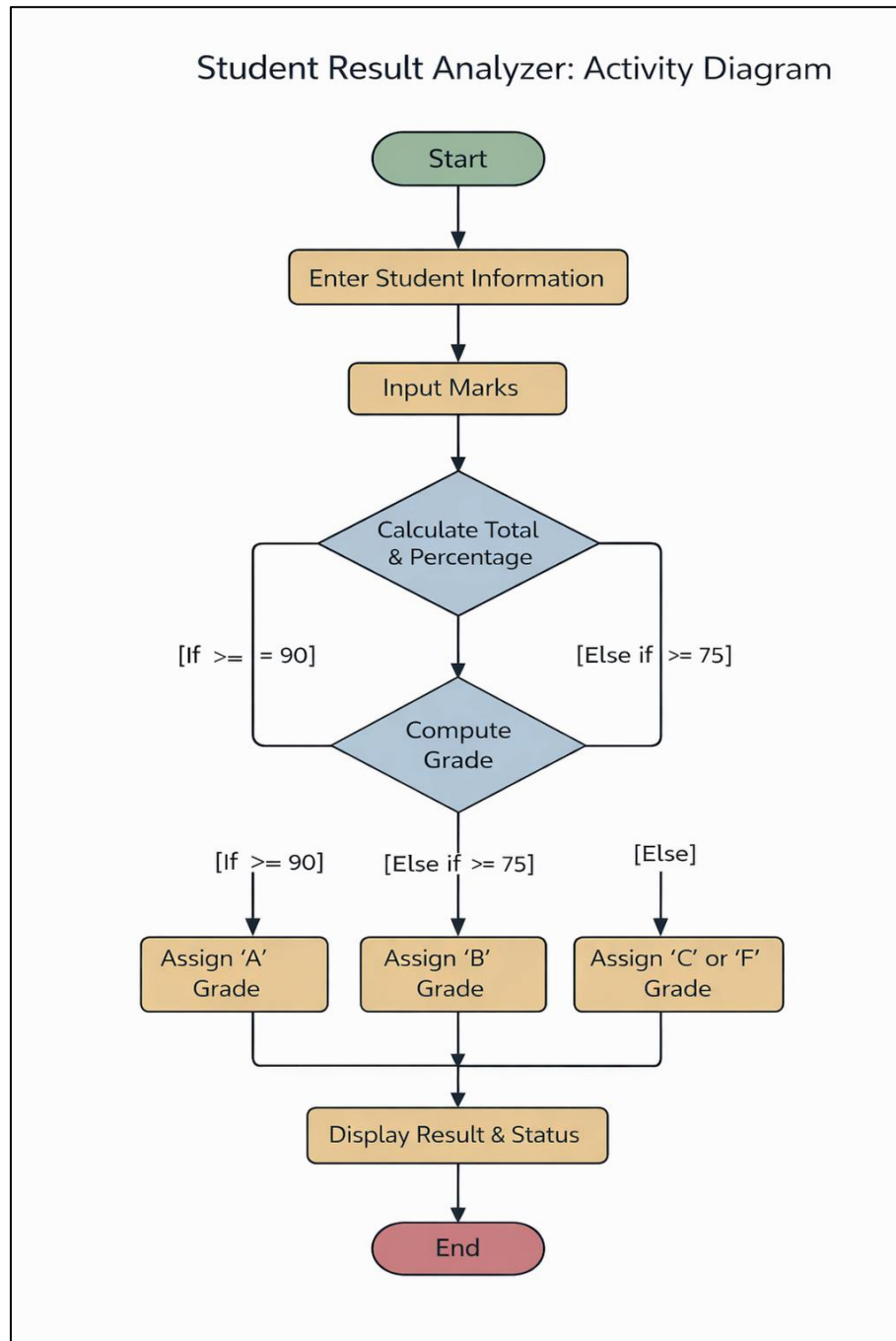
A stable view of a program is depicted in the class diagram. It reflects the many sorts of items in the system as well as their relationships. A class is made up of objects, and it may inherit from other classes. A class diagram is used to illustrate, explain, and record numerous parts of a system, as well as to create executable software code.

It displays the software system's properties, classes, functions, and relationships to provide an overview. It organizes class names, characteristics, and functions into a discrete compartment to aid in programme development. A structural diagram is a collection of classes, interfaces, linkages, collaborations, and constraints.



8.3 ACTIVITY DIAGRAM

Another important behavioural diagram in UML diagram is the activity diagram, which is used to explain dynamic characteristics of the system. An activity diagram is simply a more complex version of a flow chart that depicts the flow of information from one action to the next.



9. METHODOLOGY

The methodology explains the overall approach used for managing and analyzing student result data in a computerized system. The system accepts input from the user such as student name, roll number, and marks obtained in different subjects. After receiving the input, the program processes the data and calculates the total marks, percentage, and result status. This methodology ensures that all operations related to storing student information and analyzing results are carried out in a systematic and organized way.

The major steps followed are:

1. **Student Data Entry** – The user enters student details such as name, roll number, and marks of different subjects.
2. **Data Processing** – The system processes the entered marks and performs calculations such as total marks and percentage.
3. **Result Evaluation** – Based on the calculated marks, the system determines the pass or fail status of the student.
4. **Data Management** – The program stores and manages the student result data in an organized format.
5. **Result Display** – The system displays the final result including student details, total marks, percentage, and result status.

10. FUTURE SCOPE

The Student Result Analyzer System has good potential for future improvements and expansion. In the future, the system can be enhanced with more advanced features and technologies to make result management easier and more efficient for educational institutions. With further development, the system can become a complete academic result management solution.

The future scope of this project includes:

1. To develop a graphical user interface (GUI) to make the system more interactive and user-friendly.
2. To store student result data in a database for better data management and long-term record keeping.
3. To allow teachers or administrators to add, update, or delete student records easily.
4. To generate detailed result reports such as class performance analysis, highest marks, and average marks.
5. To integrate the system with online portals so students can view their results from anywhere.
6. To expand the system into a complete student management system including attendance, subjects, and academic records.

11. IT COMPANIES

Our project **Student Result Analyzer** can be useful for several IT and software companies that focus on developing educational software, data management systems, and digital solutions for institutions. Such companies create applications that help schools, colleges, and universities manage academic data efficiently. A system like the Student Result Analyzer demonstrates how programming can be used to automate result calculation and manage student performance records.

Many IT companies work on developing management systems for educational organizations, where accurate data handling and automation are very important. Projects like this help in understanding the basic concepts used in real-world software such as student information systems, academic management platforms, and database-driven applications.

Companies such as **Infosys Limited, Tata Consultancy Services (TCS), and Wipro Technologies** develop software solutions for educational institutions and enterprises. A project like the Student Result Analyzer reflects the basic idea of how such companies design systems that manage and process academic information efficiently using technology.

12. CODING

```
#include <iostream>

#include <fstream>

using namespace std;

class Student {

protected:

    string name;

    int roll;

public:

    void getStudentData() {

        cout << "Enter Student Name: ";

        cin.ignore();

        getline(cin, name);

        cout << "Enter Roll Number: ";

        cin >> roll;

    }

};
```

```

class Result : public Student {
private:
    int marks[5];
    int total = 0;
    float percentage;
    char grade;

public:
    void getMarks() {
        cout << "\nEnter Marks of 5 Subjects:\n";

        for(int i = 0; i < 5; i++) {
            cout << "Subject " << i + 1 << ": ";
            cin >> marks[i];
            total = total + marks[i];
        }
    }

    void calculate() {
        percentage = total / 5.0;

        if(percentage >= 90)
            grade = 'A';
    }
}

```

```

else if(percentage >= 75)

    grade = 'B';

else if(percentage >= 50)

    grade = 'C';

else

    grade = 'F';

}


void display() {

    cout << "\n----- RESULT ----- \n";

    cout << "Name: " << name << endl;

    cout << "Roll No: " << roll << endl;

    cout << "Total Marks: " << total << endl;

    cout << "Percentage: " << percentage << "%" << endl;

    cout << "Grade: " << grade << endl;


    if(grade == 'F')

        cout << "Status: Fail\n";

    else

        cout << "Status: Pass\n";

}


void saveToFile() {

```

```
ofstream file("student_result.txt", ios::app);
```

```
file << "----- RESULT -----\\n";
```

```
file << "Name: " << name << endl;
```

```
file << "Roll No: " << roll << endl;
```

```
file << "Marks: ";
```

```
for(int i = 0; i < 5; i++) {
```

```
    file << marks[i] << " ";
```

```
}
```

```
file << "\\nTotal: " << total << endl;
```

```
file << "Percentage: " << percentage << "%" << endl;
```

```
file << "Grade: " << grade << endl;
```

```
if(grade == 'F')
```

```
    file << "Status: Fail\\n";
```

```
else
```

```
    file << "Status: Pass\\n";
```

```
file << "-----\\n\\n";
```

```
file.close();
```

```
        cout << "\nData Saved in student_result.txt file";  
    }  
};  
  
int main() {  
    Result r;  
  
    r.getStudentData();  
    r.getMarks();  
    r.calculate();  
    r.display();  
    r.saveToFile();  
  
    return 0;  
}
```


13. OUTPUT

Enter Student Name: Rahul

Enter Roll Number: 23

Enter marks for 5 subjects:

Subject 1: 80

Subject 2: 75

Subject 3: 90

Subject 4: 85

Subject 5: 70

----- RESULT -----

Name: Rahul

Roll No: 23

Marks: 80 75 90 85 70

Total: 400

Percentage: 80%

Grade: B

Status: Pass

```
PS C:\Users\aditya\OneDrive\Documents\MINI PROJECT - Student Result Analyzer> g++ usingoops.cc
PS C:\Users\aditya\OneDrive\Documents\MINI PROJECT - Student Result Analyzer> ./a.exe
Enter Student Name: aditya
Enter Roll Number: 19

Enter Marks of 5 Subjects:
Subject 1: 20
Subject 2: 30
Subject 3: 40
Subject 4: 50
Subject 5: 60

----- RESULT -----
Name: ditya
Roll No: 19
Total Marks: 200
Percentage: 40%
Grade: F
Status: Fail

Data Saved in student_result.txt file
PS C:\Users\aditya\OneDrive\Documents\MINI PROJECT - Student Result Analyzer> |
```

14. BIBLIOGRAPHY

1. **Object Oriented Programming with C++** – E. Balagurusamy, Tata McGraw Hill Education.
2. **Programming in C++** – Bjarne Stroustrup, Addison-Wesley Publications.
3. **Software Engineering: A Practitioner's Approach** – Roger S. Pressman, McGraw Hill.
4. **Student Result Management System Project Reports** – Reference from academic project samples available on educational websites.
5. **Online Resources:**
 - www.geeksforgeeks.org
 - www.w3schools.com
 - www.wikipedia.org (for concept reference only)
6. **Class Notes and Practical Lab Materials** provided during the BCA course.